# Classical Machine Learning for Radar-Based Vessel Type Classification: A Comparative Study

Research Question 2: Which model delivers the classification outcome?

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**Abstract**

This paper evaluates four classical machine learning algorithms — Random Forest (RF), Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), and Gradient Boosting Trees (GBT) — for multi-class vessel type classification from terrestrial coastal radar features. Five vessel types are distinguished: Fishing (30), Dredging (33), Tug (52), Cargo (70), and Tanker (80). Using a 5-feature radar-native input set (azimuth, PeakAmplitude, TotalAmplitude, down-range extent, cross-range physical extent), we evaluate each model on accuracy, macro F1-score, area under the ROC curve (AUC), per-class recall, and prediction confidence score distributions. We characterise the inherent classification ceiling imposed by the Cargo/Tanker feature overlap (maximum Cohen's $d = 0.502$) and determine which classical algorithm best navigates this constraint.

# Contents

# 1 Introduction

Given a set of radar-derived features that describe a vessel detection, can classical machine learning reliably infer the vessel type? And if so, which algorithm is best suited to the specific structure of this problem: a 5-class imbalanced dataset where the two dominant classes (Cargo and Tanker) share nearly identical feature distributions?

This paper addresses **RQ2**: *Which model delivers the classification outcome?*

We compare four classical algorithms and answer:

1. Which algorithm achieves the highest accuracy and F1 on the balanced test set?
2. Which algorithm best handles the Cargo/Tanker confusion?
3. Do tree-based ensemble methods outperform linear and kernel methods on this feature set?
4. What confidence score distributions do different models produce, and how do these relate to prediction reliability?

# 2 Dataset and Preprocessing

## 2.1 Source Data

The dataset is the `radarfeatureL_Study.csv` collection: 123,051 radar track observations from a coastal X-band surveillance radar, fused with AIS vessel type labels (ITU-R type codes). After class filtering (retaining types 30, 33, 52, 70, 80) and removal of zero-amplitude ghost tracks, **100,430 valid observations** remain across five classes.

## 2.2 Class Distribution

Table 1: Dataset class distribution after cleaning.

| Type | Vessel | Count | Proportion |
|---|---|---:|---:|
| 30 | Fishing | 3,729 | 3.7% |
| 33 | Dredging | 6,022 | 6.0% |
| 52 | Tug | 3,540 | 3.5% |
| 70 | Cargo | 59,041 | 58.8% |
| 80 | Tanker | 28,098 | 28.0% |
| | **Total** | **100,430** | 100% |

## 2.3 Feature Set

Five radar-native features are used (see Paper 1 for full derivation):

`azimuth`, `PeakAmplitude`, `TotalAmplitude`, `down_range_extent`, `az_extent_m`

where `az_extent_m` $= r \times$ `azw` is the physical cross-range extent in metres.

### 2.4 Preprocessing Pipeline

1. **Sampling**: Up to 2,000 observations per class drawn randomly (stratified) to create a balanced 10,000-sample training pool.
2. **Train/test split**: 80/20 stratified split (training: 8,000; test: 2,000), performed *before* any oversampling.
3. **Standardisation**: StandardScaler fitted on training set only. Applied identically to test set.
4. **SMOTE**: Synthetic minority oversampling applied to training set to balance classes. Majority class count used as target for all classes.
5. **Class-weighted loss**: For MLP, class weights $w_c = N/(n_c \cdot C)$ applied to cross-entropy loss.

## 3 Classical ML Models

### 3.1 Random Forest (RF)

Random Forest [1] constructs $T$ decision trees on bootstrapped training samples, each tree using a random subset of features at each split. Prediction is by majority vote:

$$\hat{y} = \mathrm{mode}\{h_t(\boldsymbol{x})\}_{t=1}^{T} \tag{1}$$

RF provides feature importance via mean decrease in impurity (Gini importance), which is used here to rank feature contributions.

**Configuration:** 300 trees, `max_features=sqrt`, `class_weight=balanced`, random state 42.

### 3.2 Support Vector Machine (SVM)

SVM [2] solves:

$$\min_{\boldsymbol{w},b,\boldsymbol{\xi}} \frac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_i \xi_i \quad \text{s.t.} \quad y_i(\boldsymbol{w}^\top \phi(\boldsymbol{x}_i) + b) \geq 1 - \xi_i \tag{2}$$

with the RBF kernel $K(\boldsymbol{x}_i, \boldsymbol{x}_j) = \exp(-\gamma\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2)$ and one-vs-one multi-class strategy.

**Configuration:** $C = 10$, $\gamma =$`scale`, `class_weight=balanced`.

### 3.3 Multi-Layer Perceptron (MLP)

A feedforward neural network with $L$ hidden layers:

$$\boldsymbol{h}^{(l)} = \mathrm{ReLU}\Big(\mathbf{W}^{(l)}\boldsymbol{h}^{(l-1)} + \boldsymbol{b}^{(l)}\Big) \tag{3}$$

Output layer uses softmax. Trained with Adam optimiser and weighted cross-entropy loss.

**Configuration:** 2 hidden layers of width 64, Adam ($\eta = 0.001$), batch size 64, 200 epochs with early stopping (patience=20).

### 3.4 Gradient Boosting Trees (GBT)

GBT [3] builds an additive ensemble sequentially:

$$F_m(\boldsymbol{x}) = F_{m-1}(\boldsymbol{x}) + \eta \cdot h_m(\boldsymbol{x}) \tag{4}$$

where $h_m$ fits the negative gradient of the loss (pseudo-residuals). GBT can model feature interactions such as *large az_extent_m AND high TotalAmplitude* $\Rightarrow$ *Tanker*, which are invisible to single-feature statistics.

**Configuration:** 300 estimators, max depth 5, $\eta = 0.05$, subsample 0.8, sample weights proportional to class imbalance.

## 4 Experimental Setup

### 4.1 Evaluation Metrics

- **Accuracy**: $\mathrm{Acc} = \sum_c \mathrm{TP}_c / N$
- **Macro F1**: $F1_{\mathrm{macro}} = \frac{1}{C}\sum_c \frac{2\,\mathrm{TP}_c}{2\,\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}$
- **AUC**: Area under one-vs-rest ROC, averaged across classes
- **Per-class Recall**: $R_c = \mathrm{TP}_c / (\mathrm{TP}_c + \mathrm{FN}_c)$
- **Confidence**: $\mathrm{conf}(\boldsymbol{x}) = \max_c p_c(\boldsymbol{x})$

### 4.2 Baseline: Original 5-Feature Set

For reference, the original feature set (`azimuth, TotalAmplitude, ellipse_area, euclid_size, PeakAmplitude`) from the pre-study dataset (677 rows) is also evaluated. This establishes the starting point before feature engineering corrections.

# 5 Results

## 5.1 Overall Performance

Table 2: Classical ML model comparison — corrected 5-feature set, study dataset.

| Model | Accuracy | F1 Macro | AUC | Training time |
|---|---|---|---|---|
| Random Forest | 89.6% | 89.5% | 0.989 | 0.4 s |
| SVM (RBF) | 80.8% | 80.7% | 0.964 | 0.1 s |
| MLP | 77.9% | 77.6% | 0.948 | 0.2 s |
| GBT | **90.0%** | **90.0%** | 0.987 | 1.6 s |

All models trained and evaluated on the same 300-sample-per-class balanced subset (1,500 total), 80/20 stratified split (240 test rows), SMOTE applied to training set only. Training times on CPU (Intel/AMD x86-64).

## 5.2 Per-Class Recall

Table 3: Per-class recall for each classical model.

| Model | Type 30 | Type 33 | Type 52 | Type 70 | Type 80 |
|---|---|---|---|---|---|
| RF | 96% | 100% | 85% | 79% | 88% |
| SVM | 94% | 100% | 83% | 73% | 54% |
| MLP | 92% | 96% | 83% | 62% | 56% |
| GBT | **96%** | **100%** | **90%** | 77% | **88%** |

## 5.3 Confusion Matrices

The dominant confusion pattern across all models is Cargo (70) ↔ Tanker (80) misclassification. Table 4 shows the GBT confusion matrix as the best-performing model.

Table 4: GBT confusion matrix (test set, $n = 240$).

| | pred 30 | pred 33 | pred 52 | pred 70 | pred 80 |
|---|---|---|---|---|---|
| true 30 (Fishing) | **46** | 0 | 1 | 0 | 1 |
| true 33 (Dredging) | 0 | **48** | 0 | 0 | 0 |
| true 52 (Tug) | 3 | 0 | **43** | 0 | 2 |
| true 70 (Cargo) | 0 | 1 | 1 | **37** | 9 |
| true 80 (Tanker) | 1 | 0 | 0 | 5 | **42** |

Dredging (33) is perfectly classified (100% recall) across all models, confirming its distinct radar signature. Cargo (70) and Tanker (80) account for almost all errors: GBT misclassifies 9 Cargo observations as Tanker and 5 Tanker observations as Cargo.

### 5.4 Feature Importance (GBT and RF)

Table 5: Feature importance rankings from RF (Gini) and GBT.

| **Feature** | **RF Importance** | **GBT Importance** |
|---|---:|---:|
| TotalAmplitude | 0.261 | 0.254 |
| az_extent_m | 0.213 | **0.239** |
| PeakAmplitude | 0.213 | 0.171 |
| azimuth | 0.195 | 0.225 |
| down_range_extent | 0.119 | 0.112 |

### 5.5 Confidence Score Analysis

For each model, the confidence score distribution is plotted separately for correctly and incorrectly classified samples:

- A well-calibrated model shows high confidence (close to 1.0) for correct predictions and low confidence for errors.
- GBT probability outputs are often overconfident (not well-calibrated) and may benefit from Platt scaling or isotonic regression.
- RF outputs tend to be underconfident (probabilities cluster near class frequencies rather than 0 or 1).

*[Confidence histograms and calibration curves to be inserted.]*

## 6 Discussion

### 6.1 The Cargo/Tanker Ceiling

All classical models are expected to struggle with the Cargo/Tanker boundary due to the fundamental feature overlap (maximum Cohen's $d = 0.502$). The feature distributions are:

- Nearly identical mean down-range extents (293 m vs 270 m)
- Overlapping amplitude distributions
- Partially separated cross-range extents (634 m vs 898 m, $d = 0.502$)

With only single-snapshot features, the theoretical Bayes error rate is estimated to be above 20% for this class pair, setting a hard ceiling that no ML model can breach without additional features.

## 6.2 Why GBT Leads

GBT explicitly models feature interactions through decision tree splits. A split of the form “if `az_extent_m` ¿ 750 m AND `TotalAmplitude` ¿ 90000 then Tanker” cannot be captured by a linear model or even an RBF-SVM without explicit feature engineering. This interaction modelling capability makes GBT the top performer (90.0%) among classical methods on this dataset.

Both GBT and RF rank `TotalAmplitude` and `az_extent_m` as the two most discriminative features, confirming the physical intuition: vessel size (cross-range extent) and echo strength jointly determine vessel type. The physical feature `az_extent_m` — derived as range $\times$ cross-range angle width — is the only feature that achieves Cohen’s $d > 0.5$ between Cargo and Tanker ($d = 0.502$), and both tree-based methods assign it high importance.

## 6.3 Limitations of Classical Approaches

1. No explicit uncertainty modelling: classical models return point predictions; confidence scores are heuristic probabilities.

2. Feature engineering is manual: the physically derived `az_extent_m` was required to make size features meaningful.

3. Temporal features are unused: incorporating track-level statistics (SOG variability, COG steadiness) would likely improve performance.

# 7 Conclusion

This paper benchmarks four classical ML algorithms for multi-class radar vessel classification using a corrected 5-feature set derived from 100,430 clean observations (radarfeatureL_Study.csv). Key findings:

- **GBT achieves the best accuracy (90.0%, F1 90.0%, AUC 0.987)** among classical methods, marginally ahead of Random Forest (89.6%, AUC 0.989).

- **SVM and MLP underperform** (80.8% and 77.9%) primarily due to their inability to model the non-linear feature interactions that distinguish Cargo from Tanker.

- **Dredging (type 33) achieves 100% recall** in all models, confirming a distinct and separable radar signature.

- **The Cargo/Tanker boundary remains the limiting factor**: GBT achieves only 77% recall on Cargo (type 70) and 88% on Tanker (type 80). The theoretical

Bayes error from single-snapshot feature overlap ($\max d = 0.502$) prevents further improvement without temporal or multi-look features.

- **Feature importance**: both GBT and RF rank `TotalAmplitude` and `az_extent_m` highest, confirming that the physically derived cross-range extent is the most discriminative size feature for vessel type separation.

# References


[1] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

[2] Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3), 273–297.

[3] Friedman, J. H. (2001). Greedy function approximation. *Annals of Statistics*, 29(5), 1189–1232.

[4] Chawla, N. V. et al. (2002). SMOTE. *JAIR*, 16, 321–357.